

Basic Principles of Medicine 1

Module: Foundation to Medicine

DLA: Proteoglycans & Glycoproteins

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Learning Objectives: DLA: Proteoglycans and Glycoproteins

- Before lecture 'Fibrous proteins'

SOM.MK.I.BPM1.1.FTM.3.BCHM.0143	Describe the general structure of proteoglycans. Indicate sugar derivatives used for GAG synthesis.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0144	Discuss the general composition and function of GAGs. Describe the bottle-brush structure of proteoglycans.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0145	Describe the synthesis of proteoglycans, hyaluronic acid and the extracellular assembly of proteoglycan aggregates using link proteins leading to shock absorbing aggregates.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0146	Discuss the functions of hyaluronic acid and heparin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0147	Differentiate O-linked and N-linked glycoproteins and indicate the amino acids involved.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0148	Discuss blood group antigens as an example for O-linked glycoproteins.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0149	Outline synthesis of N-linked glycoproteins and indicate significance of dolichol-PP and mannose.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0150	Highlight the importance of phosphorylation of mannose (Mannose 6-P marker) in sorting of lysosomal enzymes.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0151	Predict the molecular and clinical effects of I-cell disease
SOM.MK.I.BPM1.1.FTM.3.BCHM.0152	Compare concepts of synthesis of O-linked and N-linked glycoproteins.

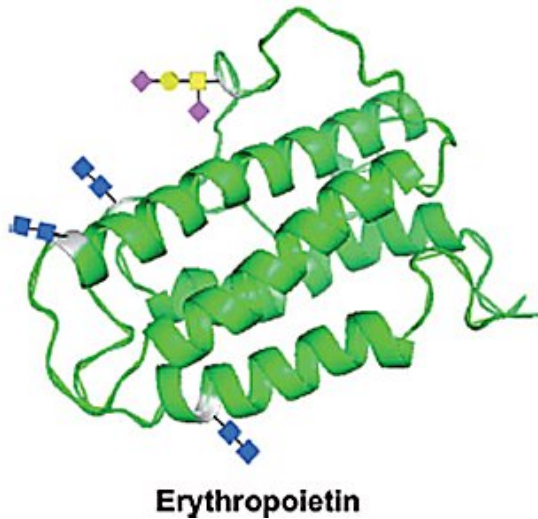
Recommended Reading

- *Lippincott's Illustrated Reviews: Biochemistry, 9th Edition:*
 -  [Lippincott Illustrated Reviews: Biochemistry, 9e](#)
- Chapter 14: Glycosaminoglycans, Proteoglycans, and Glycoproteins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
 -  [Ch 14: Glycosaminoglycans, Proteoglycans, and Glycoproteins](#)
- <https://www.clinicalkey.com/student/content/book/3-s2.0-B9780443264528000102#f0160>

Glycoproteins and Proteoglycans

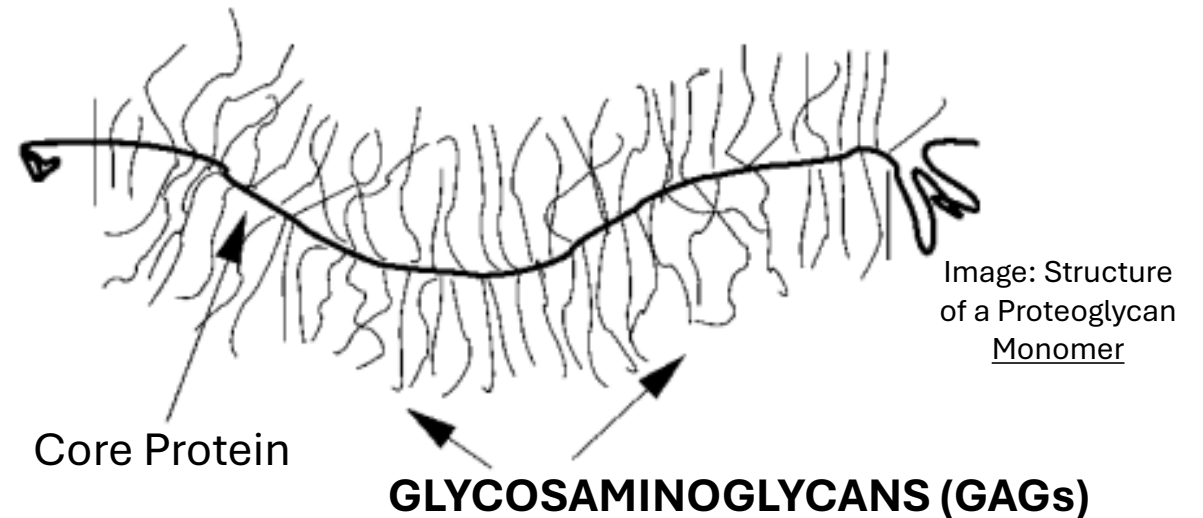
Glycoproteins

Consists mainly of **protein** + Small sugar chains (often branched).
Sugars linked via O- and N-glycosylations.



Proteoglycans

Long, unbranched polysaccharide chains (glycosaminoglycans) + Small amount of protein.

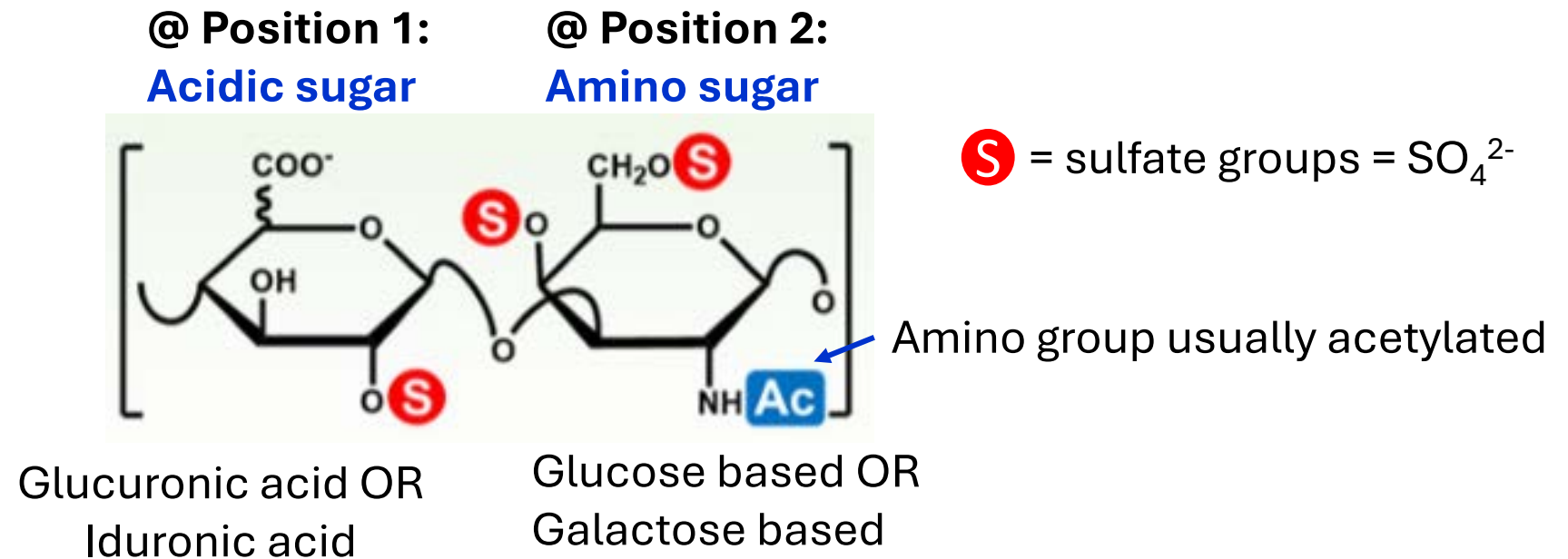


GAGs attached to protein core via O-glycosidic bonds between -OH group of serine & sugar.

Composition of Glycosaminoglycans (GAGs)

- GAGs are long repeating units of unbranched disaccharides.

General structure of repeating disaccharide units:



KEY CONCEPT: Negatively charged Sugars → GAG chains radiate out from protein core

Types of GAGs

- Permutations of these monosaccharides gives rise to different GAG families with unique properties, functions, locations and names.

Types of GAGs:

- Heparan sulfate (HS)
- Chondroitin 4- or 6-sulfates (CS)
- Dermatan sulfate (DS)
- Keratan sulfate (KS)

Conjugated to protein cores to form proteoglycans

- Hyaluronan/ hyaluronic acid (HA)
- Heparin (Hep)

Free GAGs

- Located throughout ECM & on cell membranes in all tissues (**Extracellular**)
- Heparin: **Intracellular** (within mast cells)

Structure of Proteoglycan Monomer

- Proteoglycan contains > 100 GAG chains
- A single GAG chain can contain up to 200 disaccharide units
- GAGs are covalently linked to serine residues of core protein via **O-glycosylation**
- The GAG chains on a single monomer can differ (see example below)

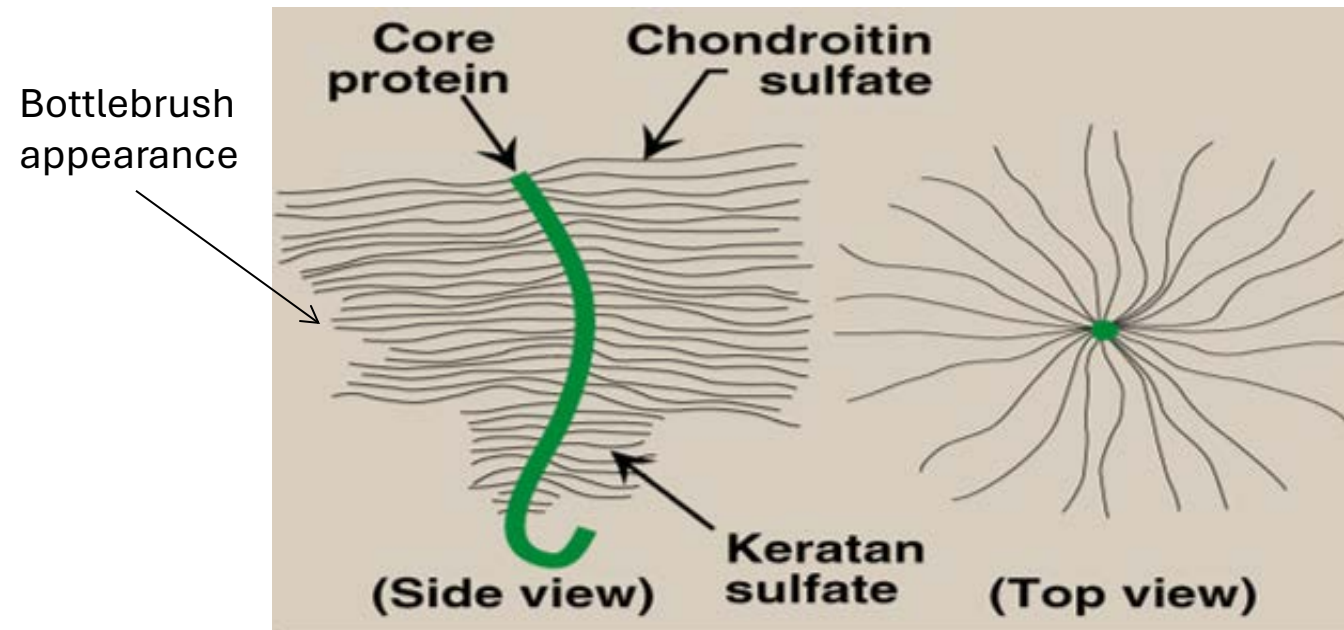


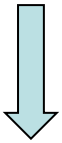
Image: Cartilage proteoglycan which contains mainly chondroitin sulfate and keratan sulfate

Structure and Function of Proteoglycans

High number of negative charges on GAGs causes:

- Extension of each chain
- Repulsion between chains (bottle-brush appearance)
- **Binds water!**

Generates a **hydrated gel**



Flexible support of the ECM
Tissue hydration
Lubrication
Shock absorption
Molecular sieve

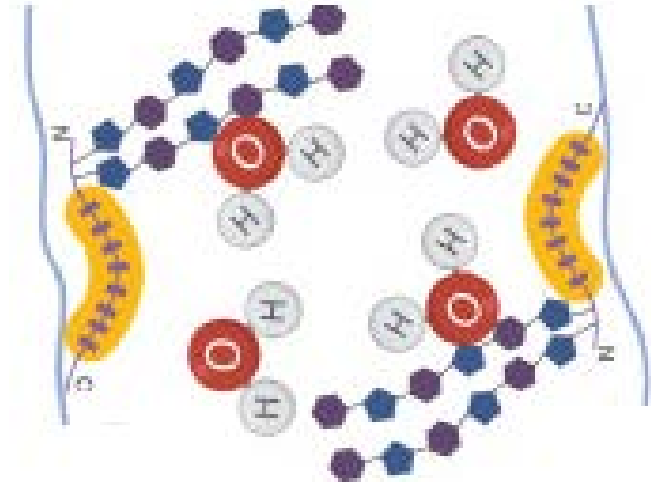
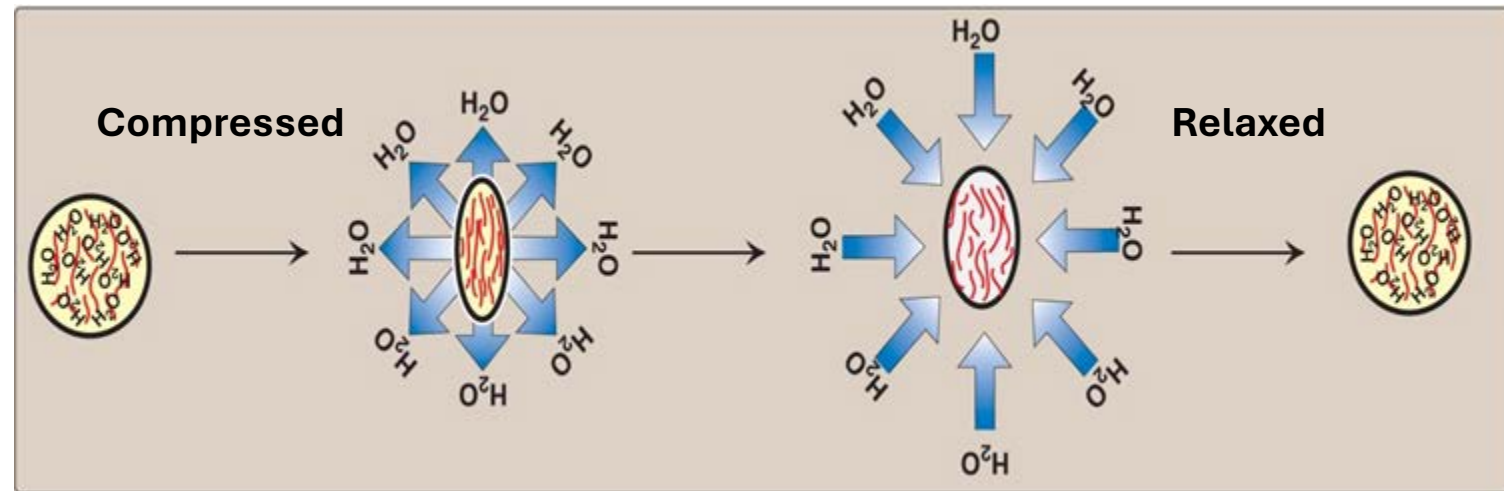


Image adapted from Hua R *et al.*,
2020. Matrix biology 94, 95-109



Resisting Compressive Forces

Functions of GAGs on Cell Surfaces

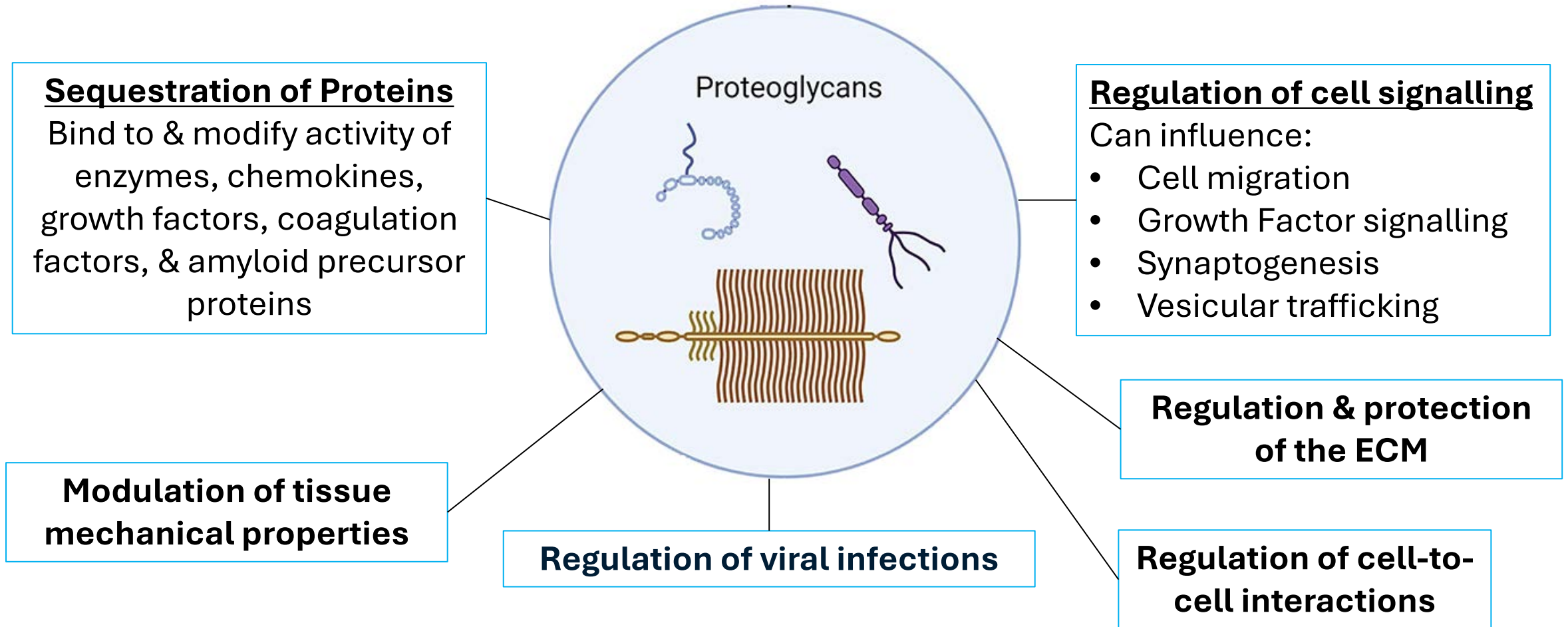
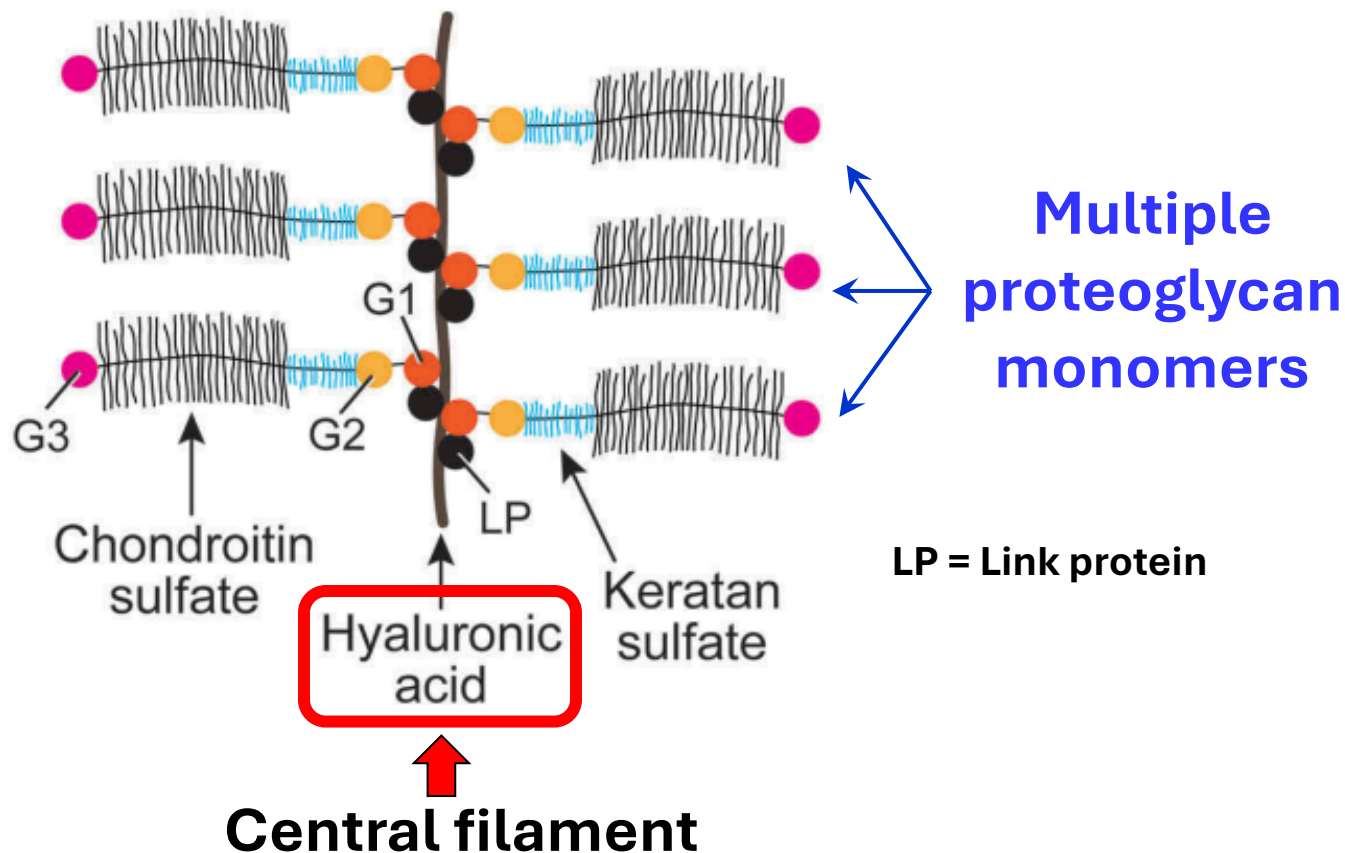


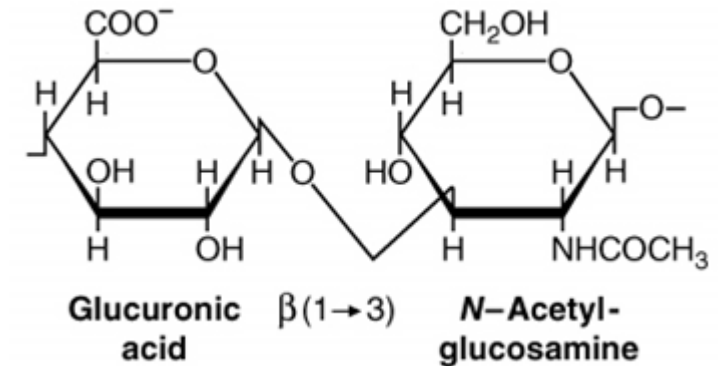
Image adapted from Nguyen & Panitch. Am. J
Physiol Cell Physiol 322: C754–C761, 2022

Proteoglycan Aggregates: Hyaluronic acid + Proteoglycan monomers

- **Proteoglycan monomers** associate with 1 molecule of **hyaluronic acid (hyaluronan)** to form **proteoglycan aggregates**.



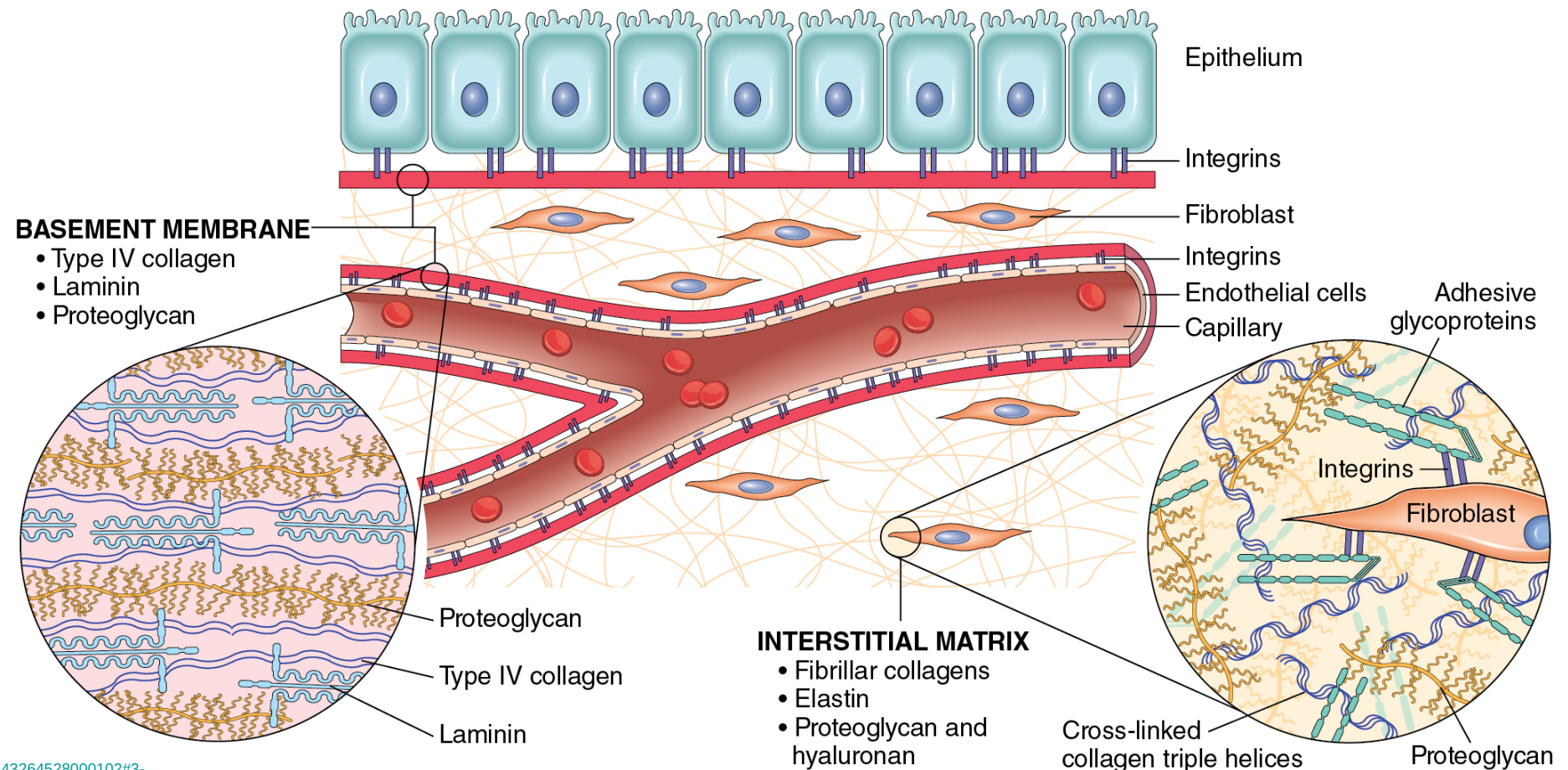
Hyaluronic acid (HA)



- **Not sulfated**
- HA connected to core proteins of proteoglycans **via link proteins**
- Synthesized directly into the ECM

Proteoglycan Aggregates in Extracellular Matrix

- Interstitium (ECM) and basement membrane: Contain structural proteins (collagen/ elastin) + Ground substance
- Ground substance: Proteoglycan aggregates + Multi-adhesive Glycoproteins (fibronectin/ laminin)

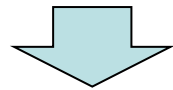


Functions of Hyaluronic Acid (HA)

- **Proteoglycan aggregates (HA + proteoglycan monomers)** → contribute to the structural and mechanical properties of tissues.

- **Functions of HA alone:**

- Hydration
- Lubrication of joints
- Space filling capacity
- Framework through which cells migrate

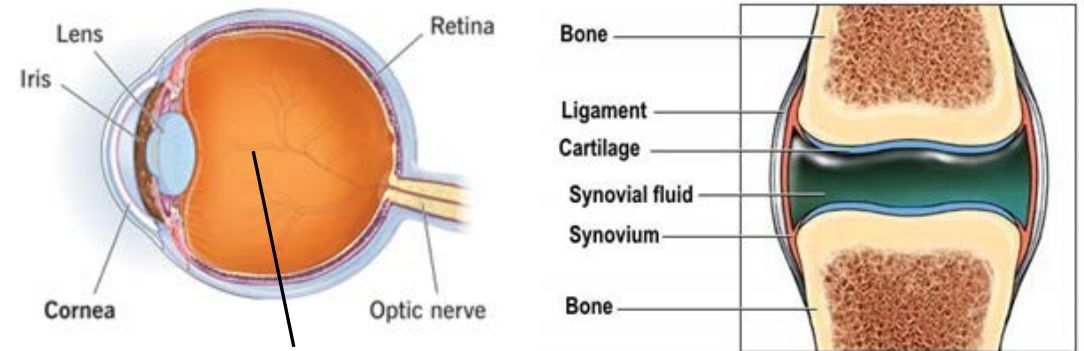


Facilitates cell migration during:

1. Embryogenesis (Neural tube closure)
2. Morphogenesis
3. Wound repair

Found in:

1. Vitreous humor of eye.
2. Synovial fluid of joints.
3. Cartilage.
4. Loose connective tissues.



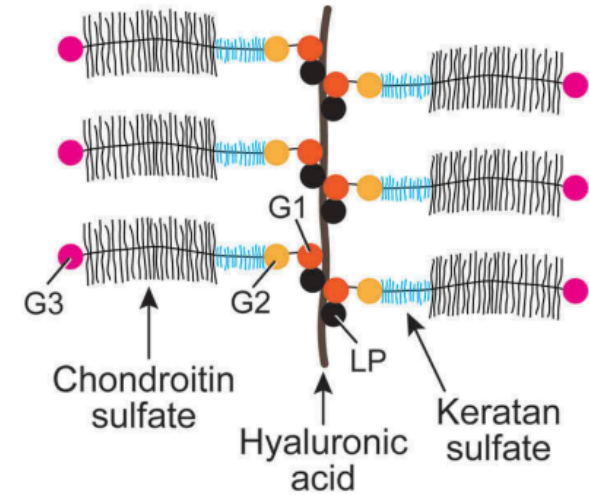
Vitreous humor:

98% water

2% hyaluronic acid

Chondroitin Sulfate: Most Abundant GAG

- **Disaccharide unit** = acidic sugar + N-acetylated amino sugar
- Found in bone, cartilage, tendons, ligament, and aorta
- Form proteoglycan aggregates through non-covalent association with hyaluronic acid
- Dietary supplementation: Osteoarthritis treatment



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Glucosamine and Chondroitin for Osteoarthritis: What You Need To Know

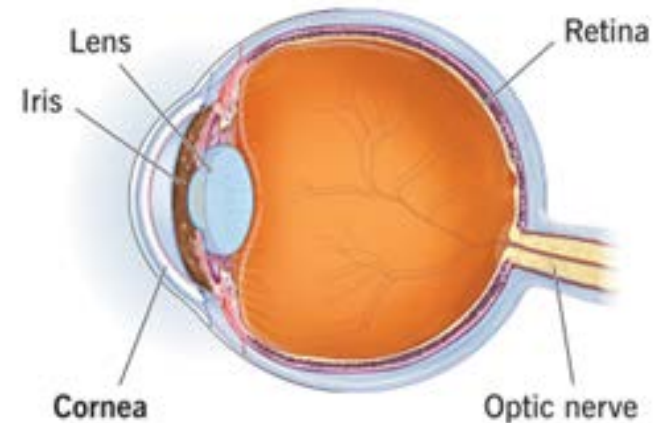
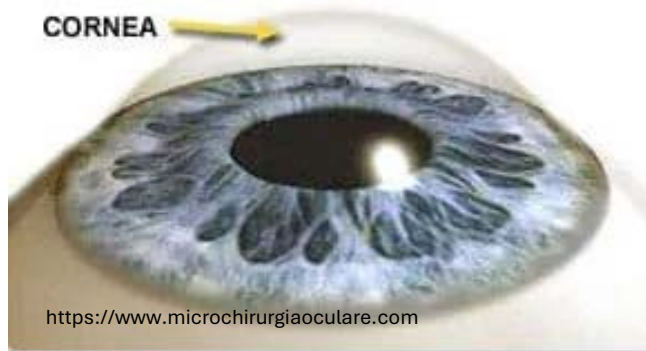
What is osteoarthritis?

Osteoarthritis is a degenerative joint disease in which cartilage and other tissues within the joint break down or change in structure. Symptoms can include pain, joint stiffness, swelling, and difficulty in moving the joint. The joints most frequently affected by osteoarthritis include the knees, hips, and hands.



Keratan Sulfate

- **Disaccharide unit = Different!** → galactose (posn 1) + N-acetylated amino sugar (posn 2). **S** may be present on sugars.
- Most **heterogeneous GAG**; May contain additional monosaccharides.
- Abundant in Cornea (for transparency).
- Also found in connective tissue proteoglycan aggregates with chondroitin sulfate (slide 8).



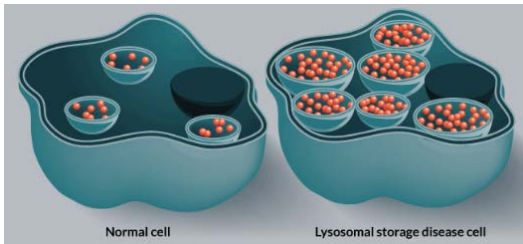
<https://my.clevelandclinic.org>

Dermatan Sulfate

- **Disaccharide unit** = acidic sugar + N-acetylated amino sugar
- Abundant in skin, cartilage, blood vessels, and heart valves

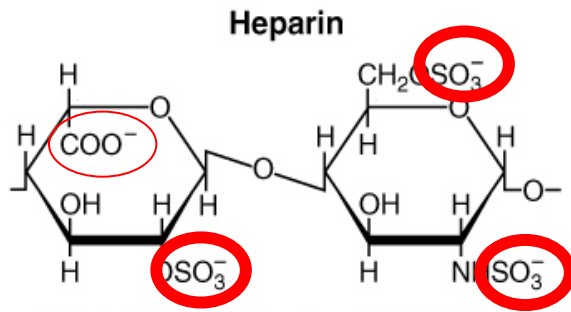
Heparan Sulfate

- **Disaccharide unit** = acidic sugar + N-acetylated amino sugar
- Found in basement membranes and common on cell surfaces
→ cell-cell recognition



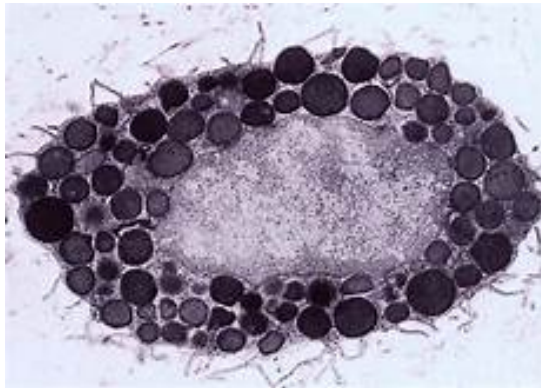
Mucopolysaccharidoses

- ❖ **Lysosomal** degradation of GAGs requires specific acid hydrolases
- ❖ **Mucopolysaccharidoses (MPS)**: Inherited Deficiency in one of the acid hydrolases
 - ❖ Hunter and Hurler Syndrome: Accumulation of **dermatan and heparan sulfate**
- ❖ Lysosomal GAG accumulation → Skeletal deformities, hepatomegaly, and intellectual disability



Heparin

Disaccharide unit = acidic sugar + amino sugar.
Highly **sulfated** and **negatively charged**



Intercellular granules of mast cell containing heparin

- **INTRACELLULAR**
- In **mast cell granules** in arteries of lung, liver, and spleen.
- Acts as **anticoagulant** (inhibits thrombin and prevents clotting)

Anticoagulant: Prevents and treats thrombotic events

- Deep vein thrombosis (DVT)
- Pulmonary embolism (PE)
- Atrial fibrillation (AF)



Summary of GAGs and Proteoglycans: Structure-Function

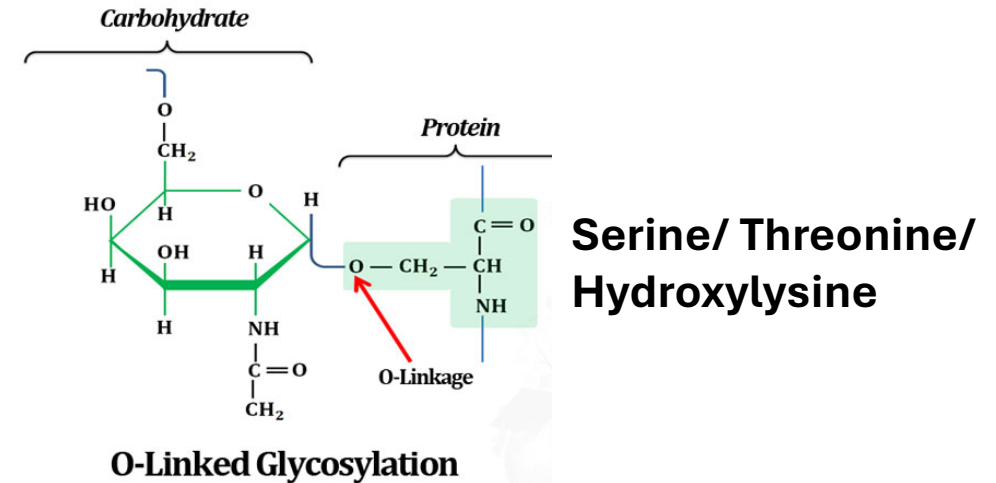
Type of GAG	Repeating disaccharide of acidic sugar (posn. 1) and N-acetylated amino sugar (posn. 2)	Sulfated	Extra-cellular	O-linked to protein	Special function	Disease related
Chondroitin sulfate	✓	✓	✓	✓	Most abundant; Cartilage	Treatment for Osteoarthritis
Dermatan sulfate	✓	✓	✓	✓	Lysosomal degradation defect	MPS: Hunter, Hurler syndromes
Keratan sulfate	✗ galactose (posn.1) + N-acetylated amino sugar (posn. 2)	✓	✓	✓	Transparency of cornea	
Heparan sulfate	✓	✓	✓	✓	Lysosomal degradation defect	MPS: Hunter, Hurler, syndromes
Heparin	✓	✓	✗ In mast cells (Intracellular)	✗	Anticoagulant	
Hyaluronic acid	✓	✗	✓	✗	Cell migration, Central strand of proteoglycan aggregates	Potential as a biochemical marker for cancer

Glycoproteins

- Carbohydrates attached to proteins via **O-linked** or **N-linked glycosylation**.

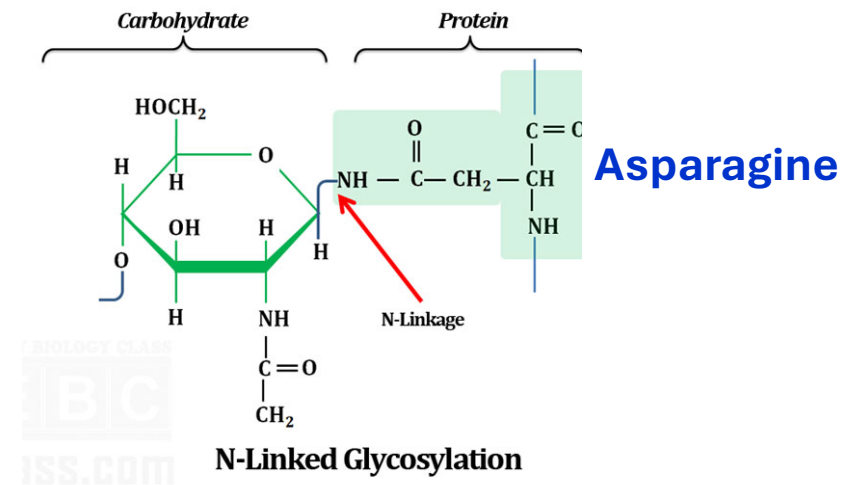
❖ O-linked glycosylation:

Sugars attached via **OH** groups of **Serine** or **Threonine** residues OR **Hydroxy-lysine** residues (e.g., collagen – FTM 28)

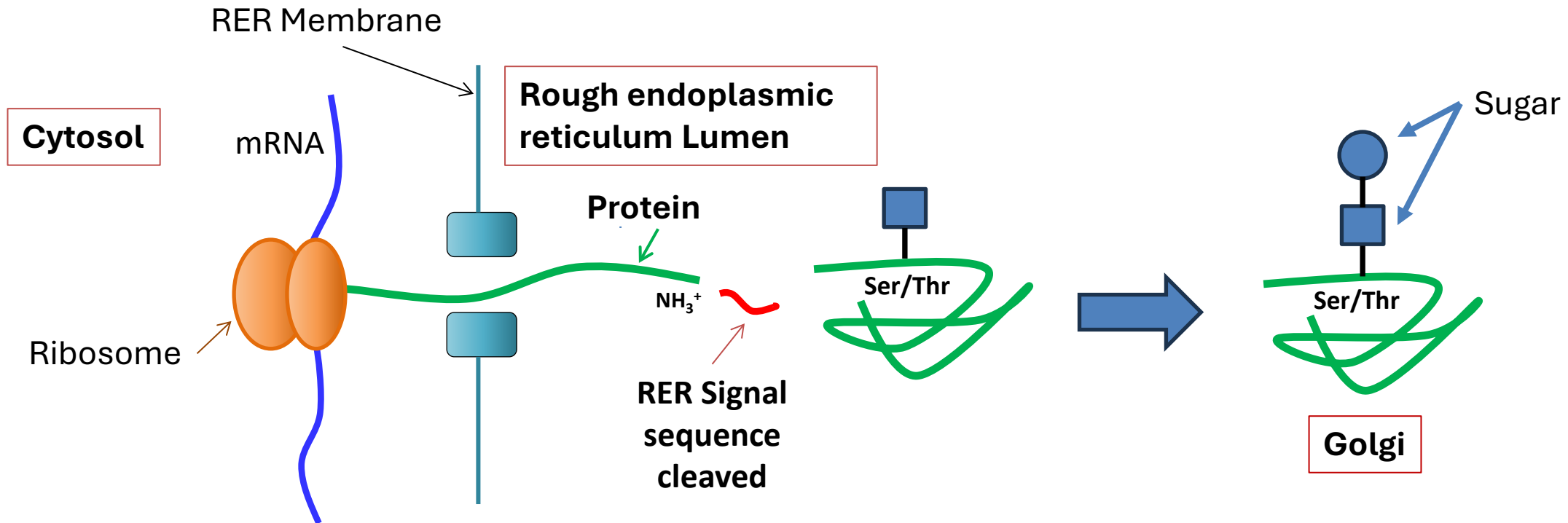


❖ N-linked glycosylation:

Sugars are attached via **amide N** group on side chain of **Asparagine** residues



O-Glycosylation of Proteins

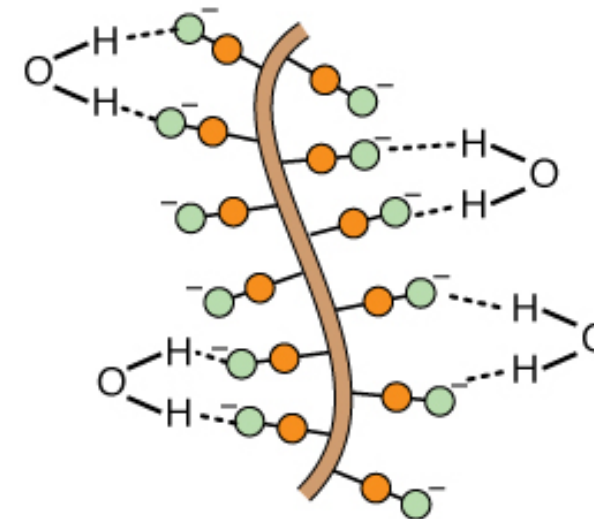
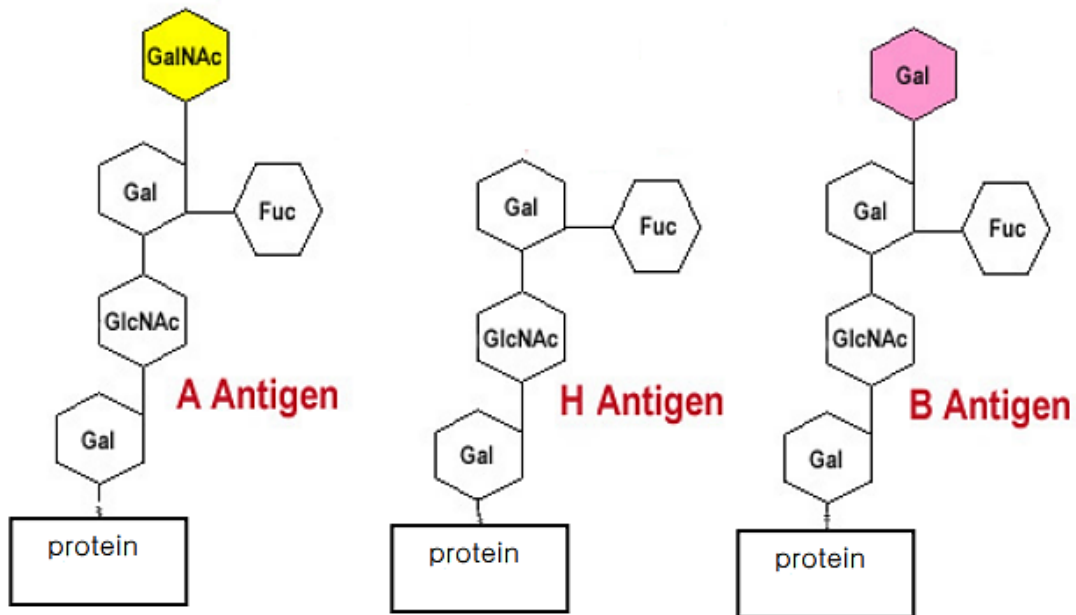


- In the ER → 1st sugar enzymatically linked directly to the OH-group of serine, threonine or hydroxylysine by **Glycosyltransferases**.
- ❖ Subsequent addition of individual sugars occurs in **Golgi**. Golgi membranes contain glycosyltransferases.

Examples of O-Glycosylation

Examples of O-linked glycoproteins: Proteoglycans, collagen, glycocalyx components, **blood group** substances, and **mucins**.

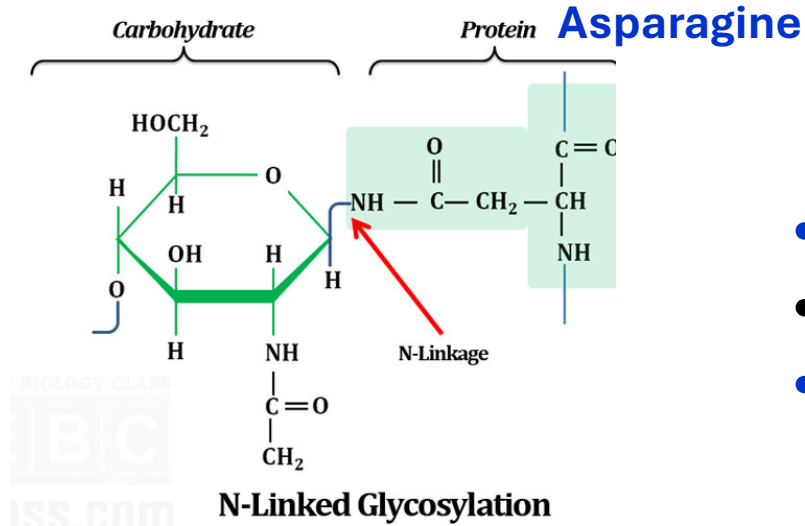
Blood Group Substances



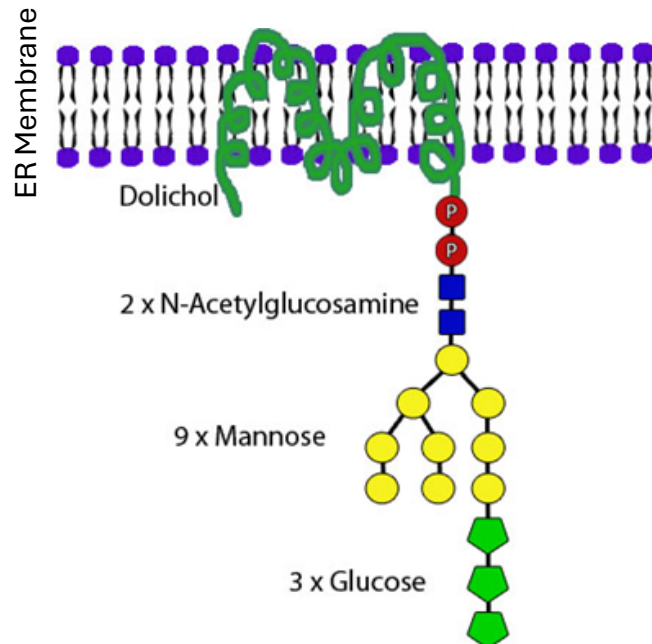
Salivary mucin

● = Sialic acid
● = N-acetylglucosamine

N-linked Glycoproteins



- **Only Asparagine residues** are N-glycosylated
- N-glycosylation requires **dolichol-pyrophosphate**
- **A mannose-rich oligosaccharide** is synthesized bound to the lipid **dolichol-PP**
- Biological functions of N-linked glycoproteins:
 - Protein folding
 - Protein trafficking
 - Signal transduction



N-Glycosylation of Proteins

- RER: **Mannose-rich oligosaccharide** transferred to specific **asparagine** residue of protein to be N-glycosylated (point 3)
- Trimming of sugar residues RER (point 4)
- Further modifications in Golgi (point 5)

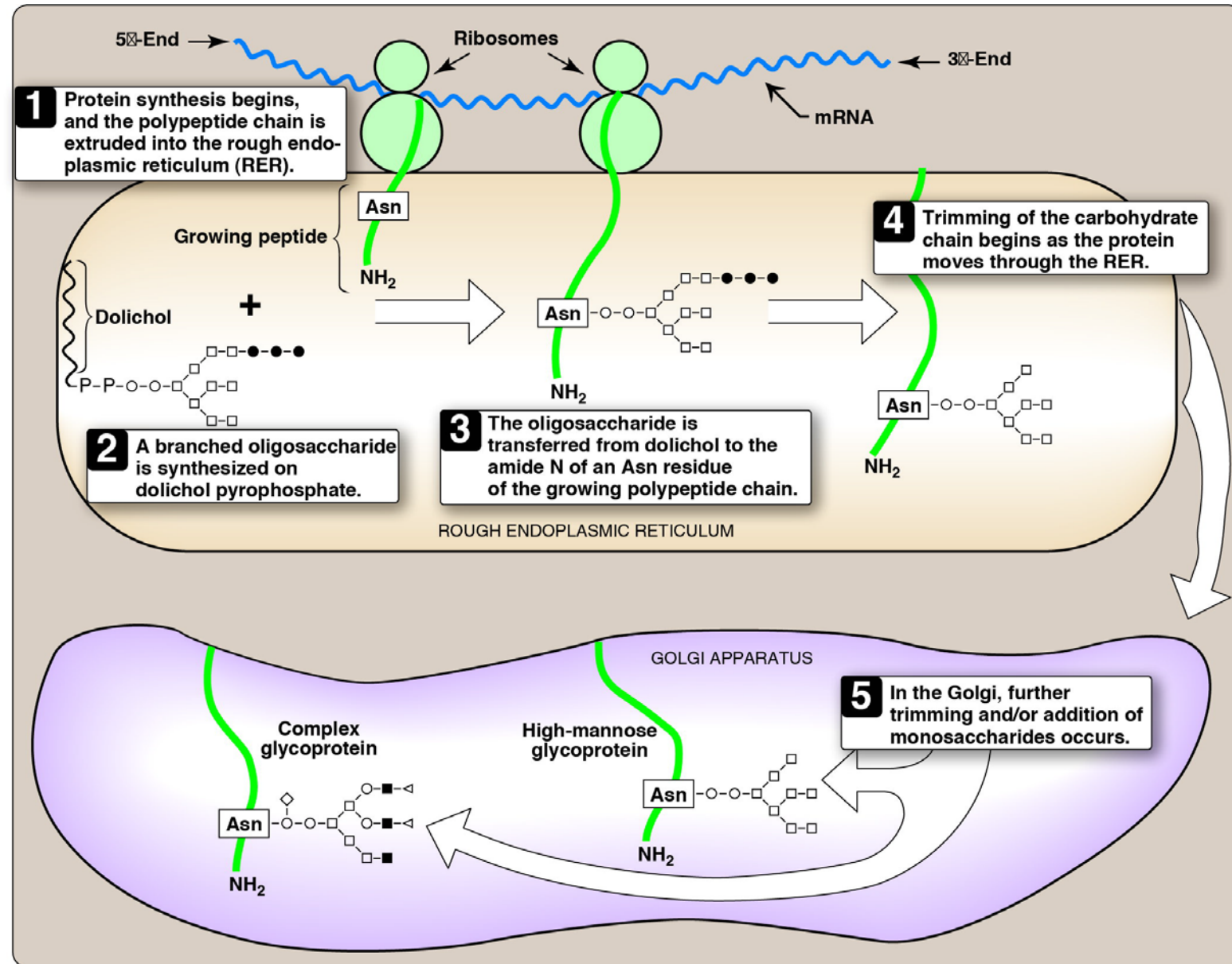
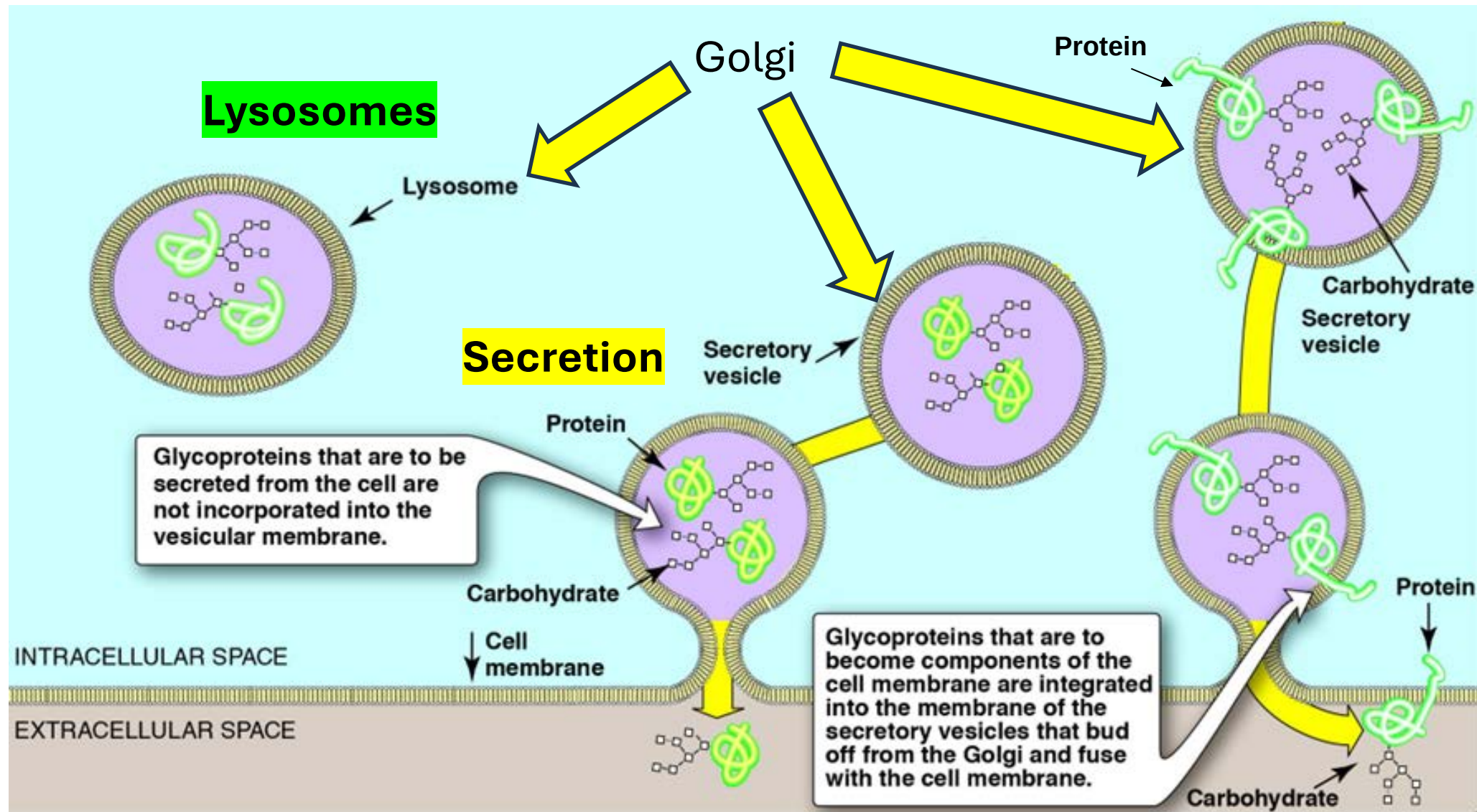
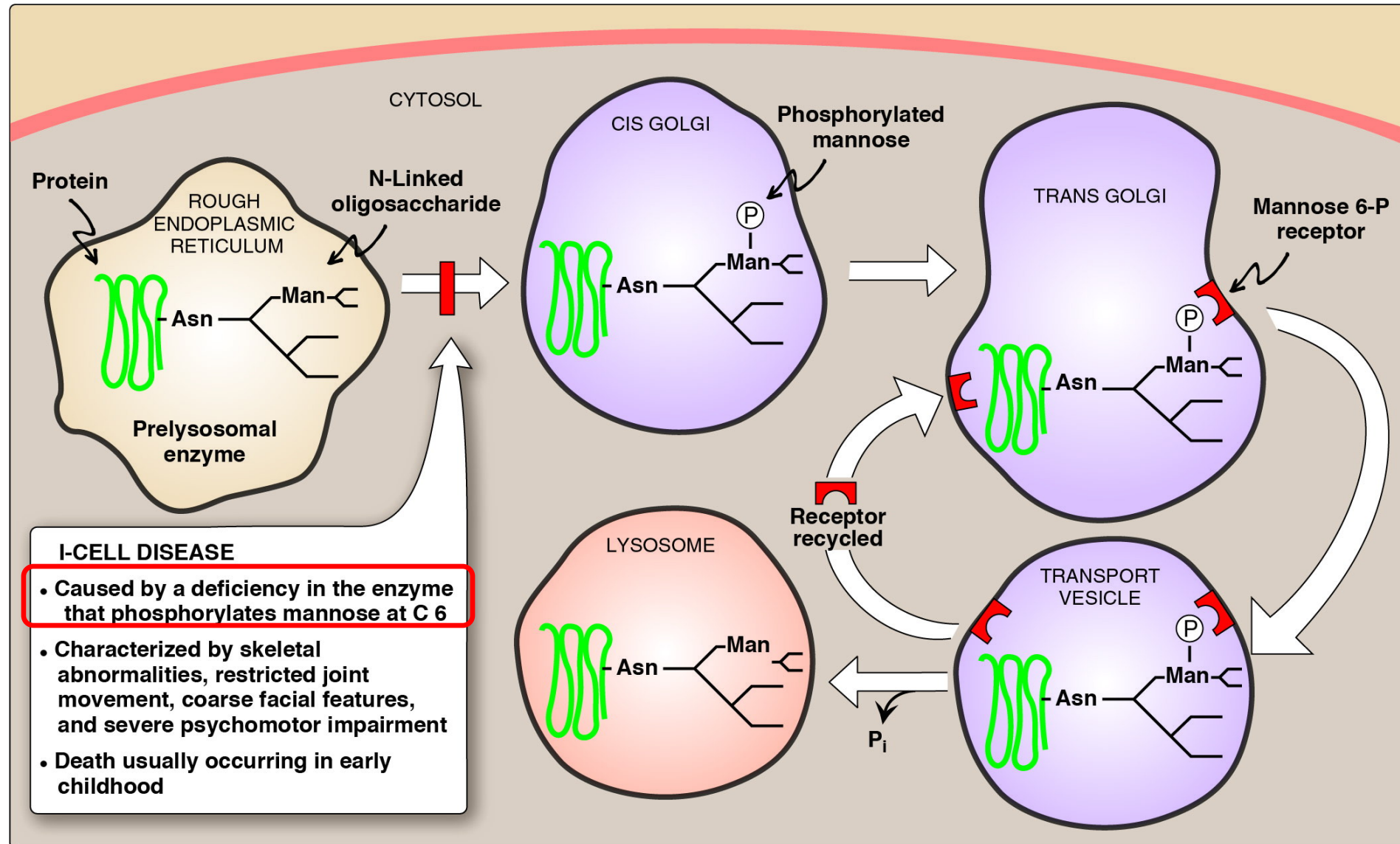


Figure 14.16, Lippincott 9e

Fates of N-linked Glycoproteins: **Secretion**, **Lysosomes**, **Cell Membrane Proteins**



Enzymes Destined for Lysosomes Require Addition of **Mannose 6-Phosphate** Marker for Transport



Transport of lysosomal enzymes: Role of **Mannose 6-phosphate**

- **Lysosomal enzymes are N-linked glycoproteins**
- They receive **mannose 6-P marker** (cis-Golgi) for transport into **lysosomes**.
- **Mannose 6-P marker is formed** by a **specific phosphotransferase** that recognizes all **enzymes** destined to be **transported** into **lysosomes**.
- **Mannose 6-P receptors** (trans-Golgi) bind the marked **enzymes** and **package** them into **vesicles** for **transport** into **lysosomes**
 - **Sorting and trafficking into lysosomes**
- **Deficiency of the mannose 6-P marker leads to I-Cell disease**
 - The potential **lysosomal enzymes** are **lacking** in lysosomes and are instead **secreted** and **found** in **plasma** and **urine**.
 - The disease manifests at birth and often has a **fatal outcome in early childhood** due to **cardiopulmonary complications**



<https://emedicine.medscape.com/article/945460-overview?form=fpf>

I-Cell disease: Lysosomal Storage Disease

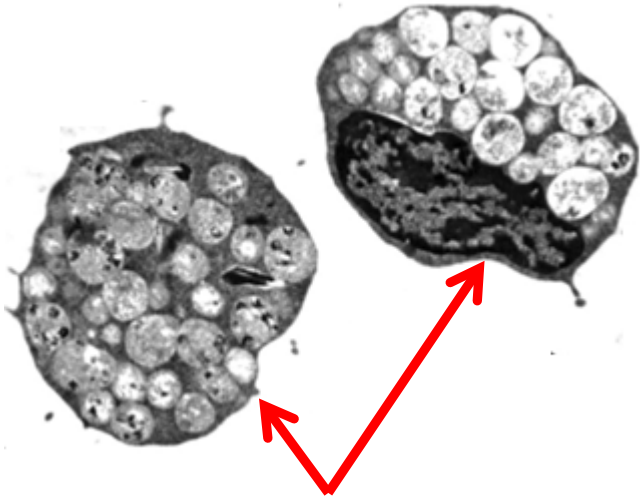


Image: Peripheral blood lymphocytes
with vacuole-like inclusions
jcp.bmj.com

- ❖ Lysosomes lack several enzymes
- ❖ Enzymes destined for lysosomes are instead secreted into the bloodstream.
- ❖ As a result, **many macromolecules** are **not degraded** and **accumulate** forming **inclusion bodies**.
 - ❖ **Complex lipids and GAGs accumulate**

- **Diagnosis:** Inclusions in lymphocytes (easily obtained) or cultured skin fibroblasts; And confirmed by **lysosomal enzyme activity** in blood
- Clinical features similar to mucopolysaccharidoses or sphingolipidoses
- I-Cell disease: Also known as mucopolipidosis Type II.

Summary & Comparison

O-glycosylation

Individual linkage of sugars step by step directly to protein.

- First **sugar** is linked in the **RER**. **Further modifications of the sugar chain** in **Golgi**. The **oligosaccharides** are often **branched**.
- **First sugar** is linked to **-OH-group** of **serine** or **threonine** residue of **protein**. (In **collagen** to **hydroxylysine residues**)
- Examples: collagen, proteoglycans, mucins, glycoproteins, and blood group substances.

N-glycosylation

Formation of mannose-rich oligosaccharide bound to the lipid dolichol pyrophosphate.

- The **mannose-rich precursor** is linked to the **nitrogen** of an **asparagine** side chain in protein.
- Later, sugars are modified in the RER and Golgi, depending on the **protein**.
- **O-linked glycoproteins are formed in the Golgi.**
- **Fate of N-linked glycoproteins Golgi:** Secretory proteins, cell membrane proteins, or transport into lysosomes.
- Transport into **lysosomes** requires mannose 6-P marker, and if deficient, leads to **I-cell disease**.

Thank you!

Please email kpadget@sgu.edu if you have questions.